

Figure 1: Normal distribution with mean = 0 and std = 3

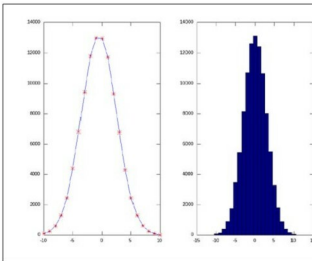


Figure 2: Histograms of normal distributed random points

```

2.32727
-5.37929
-0.63447
octave:2> mean(Y)
ans = 0.042839
octave:3> mean(Y, "h")
ans = -4.3226
octave:4> std(Y)
ans = 3.8662
octave:5> median(Y)
ans = -1.0442
octave:6> cov(Y, Y)
ans = 14.947
octave:7> cor(Y, Y)
ans = 1
octave:8>

```

Among the many moments, the four common ones are: 1) *median()* gets the middlemost element in the sorted

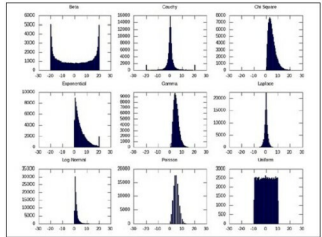


Figure 3: Histograms of probability density functions

arrangement of data; 2) *mode()* gets the most frequently occurring data point; 3) *cov()* gives the variance between two sets of data points, i.e., the covariance; and 4) *cor()* gives the relation between two sets of data points, i.e., the correlation ranging from -1 to 1. A correlation of 1 indicates they are completely related, 0 indicates they are totally unrelated, and -1 indicates they are completely related, but inversely.

Visualising the probabilities

Random numbers are a beautiful example of probabilities. They occur as per their probabilities, decided by the probability density function (PDF) they follow. Let's visualise that, using a live example. Again, as in the above example, let's look at some random numbers following the normal distribution. But, this time we should take a lot more points, say 100,000 points, so that we can actually see them following the bell-shape. But still, we are not having infinite points to give the continuous bell. So, what we have to do is collect the random points around some pre-designated buckets of fixed ranges. That is what a histogram is. *histc()* does exactly that, returning the number of points in each of the buckets, which we can then plot to see our bell. Another function, *hist()*, does all those in a more beautiful way. Figure 2 shows the two plots, for which the code is as follows:

```

$ octave -qf
octave:1> Y=normrnd(0, 3, 100000, 1); # 1 lakh random points
octave:2> B=-10:1:10; # Bucket edges -10, -9, ..., 10, i.e.
21 buckets
octave:3> N=histc(Y, B);
octave:4> subplot(1, 2, 1);
octave:5> plot(B, N, "r+", B, N, "b-");
octave:6> subplot(1, 2, 2);
octave:7> hist(Y, B); # More beautiful boxy plot
octave:8>

```

Similarly, if we use other PDFs, we would see histograms